

Python Code

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1 #####
2 #             Trie                                     #
3 #####
4
5 class TrieNode():
6     def __init__(self):
7         self.childnode = {
8             'a': None, 'b': None, 'c': None, 'd': None, 'e': None, 'f': None,
9             'g': None, 'h': None, 'i': None, 'j': None, 'k': None, 'l': None,
10            'm': None, 'n': None, 'o': None, 'p': None, 'q': None, 'r': None,
11            's': None, 't': None, 'u': None, 'v': None, 'w': None, 'x': None,
12            'y': None, 'z': None
13        }
14        self.is_word = False
15
16 class Trie():
17     def __init__(self):
18         self.root = TrieNode()
19
20     def insert(self, word):
21         temp = self.root
22         for char in word:
23             if temp.childnode[char] is None:
24                 temp.childnode[char] = TrieNode()
25             temp = temp.childnode[char]
26         temp.is_word = True
27
28     def search(self, word):
29         temp = self.root
30         for char in word :
31             # if char not in temp.childnode:
32             if temp.childnode[char] is None:
33                 return False
34             temp = temp.childnode[char]
35         return temp.is_word
36     def delete(self, word):
37         temp = self.root
38         for char in word :
39             # if char not in temp.childnode:
40             if temp.childnode[char] is None:
41                 return
42             temp = temp.childnode[char]
43         temp.is_word = False
44
45 # Sample Usage
46 trie = Trie()
47 # Insert words into the trie
48 trie.insert("apple")
49 trie.insert("app")
50 trie.insert("apricot")
51 trie.insert("banana")
52 # Search for words in the trie

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53 print(trie.search("apple"))
54 print(trie.search("app"))
55 # Output: True
56 # Output: True
57 print(trie.search("apricot")) # Output: True
58 print(trie.search("banana")) # Output: True
59 print(trie.search("orange")) # Output: False
60 trie.delete("app")
61 print(trie.search("app"))
62 # Grading Criteria
63 # Output: False
64
65
66
```